

YONGSHENG MEI

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SUMMARY: Applied Scientist developing the Large Language Models (LLMs) powering the voice assistant Alexa+, significantly improving its agentic capability and enabling its successful launch in 2025, with experience includes **Foundation Models & LLM Post-Training, Reinforcement Learning (RL) from Human/AI Feedback, LLM Agent Harness & Agentic RL**.

SKILL

Programming Language	Python, C, C++, Java, Go, SQL, MATLAB, R
Library/Framework	PyTorch, VERL, Megatron, DeepSpeed, NeMo, MCP, Hugging Face, vLLM, Ray, SkyRL, Scikit-learn
Tool	Git, Jupyter, $\LaTeX$, Kubernetes, Docker, Spark, MLflow, Cline, Claude Code, Codex
Database	MySQL, PostgreSQL, MongoDB, ChromaDB

EDUCATION

The George Washington University <i>PhD in Electrical Engineering</i> Research Areas: Reinforcement Learning, Generative Models, Bayesian Optimization	Sept. 2019 – Oct. 2024 <i>Washington, DC, US</i> GPA: 4.00
Huazhong University of Science and Technology <i>BE in Automation Engineering</i> Relevant Courses: Pattern Recognition, Control Theory, Computer Architecture	Sept. 2015 – June 2019 <i>Wuhan, Hubei, China</i> GPA: 3.81

EXPERIENCE

Applied Scientist <i>Amazon, Alexa AI</i>	Nov. 2024 - Present <i>Bellevue, WA, US</i>
- Developed Alexa's central foundational LLMs empowering the public launch of Alexa+, an intuitive, intelligent, and responsive conversational assistant, increasing the capabilities built into the model and enabling the use of thousands of APIs and external knowledge sources to serve the best experience for each request across millions of customers and endpoints. - Owned LLM post-training (SFT/RL/DPO) and evaluation for dense/MoE models. As validated by 100+ Alexa+ benchmark metrics, the models outperformed leading external models (e.g., Claude Sonnet 4.5/4/3.5) on accuracy, latency, and capacity for over 90% of real-user traffic on Alexa+. - Owned end-to-end RL training and reward design across successive production model versions, conducting ablations to select the reasoning RL algorithm for MoE models and delivering the final checkpoint. Drove gains across instruction-following, API accuracy, and temporal reasoning (+5% to a new best) while controlling chain-of-thought length to cut inference latency. - Designed and built an agentic workflow for a device-control domain, a model-agnostic harness that improves action accuracy without retraining. Evolved the design across three generations: from a deterministic guidance-injection harness to a standardized MCP server and a generic dispatch architecture supporting arbitrary third-party APIs, lifting the failure-set pass rate from a 25.6% LLM-only baseline to 84.7%, and built K8s distributed evaluation that cut full-set eval turnaround from ~9 hours to ~50 minutes (10$\times$). - Built and tuned an LLM-based guardrail safety layer routing traffic between models to prevent customer-experience regressions, reaching up to 97% blocking accuracy while driving false-blocking rates below 5%, enabling more traffic to shift to the primary model for lower latency and cost.	
Research Assistant <i>The George Washington University</i>	Sept. 2019 - Oct. 2024 <i>Washington, DC, US</i>
Reinforcement Learning: Developed an optimized collective prioritized experience replay scheme for off-policy multi-agent RL (outperforming baselines by up to 10%), and applied system-algorithm co-design for RL acceleration via a data-reuse strategy (25.4% training-time reduction for 32 agents) and a virtual cache for offline RL sampling (6$\times$ speed-ups). Stochastic Modeling & Bayesian Optimization (BO): Developed a gradient-aware BO framework to find local optima in multimodal functions for hyperparameter tuning, and a Gaussian Cox process BO model on spatiotemporal data to estimate arrival intensity (outperforming baselines in 7 of 9 settings) and detect regions of interest. Multimodal Learning: Designed a multimodal brain-tumor MRI segmentation model that improves accuracy via self-attention over correlated common information microstructures across modalities, reaching 92% whole-tumor accuracy on BraTS-2020.	

- **Network Security:** Led a reliable **self-synchronizing moving target defense** via customized network/IoT protocols (defending MITM and DoS), and introduced a **real-time network intrusion detection** model over arriving-packet sequences using a causal decision transformer.

Research Engineer Intern

Intelligent Fusion Technology, Inc.

May 2024 – July 2024

Germantown, MD, US

- Designed and applied the **transfer learning** algorithm to the **domain adaptation** task to improve the performance of civilian and military vehicle recognition for a US defense program.

Visiting Researcher

Purdue University

June 2023 – Aug. 2023

West Lafayette, IN, US

- Developed a **continual federated learning** model with time-variant input of each edge device, which leverages the **conditioned diffusion model** that is trained from scratch and fine-tuned from a pre-trained model to generate synthetic data to avoid the catastrophic forgetting problem.

PUBLICATION

1. Jingquan Wang, Jun Yin, Xu Han, **Yongsheng Mei**, Jie Hao, Bin Guo, *PB-GRPO: Learning Socially Adaptive LLM Agents from Persona-Driven Simulation with Preference-Batched GRPO*, Annual Conference on Language Modeling (COLM) Social Sim workshop, 2026.
2. Yongchang Hao, Jie Hao, **Yongsheng Mei**, Ze Ye, Junyi Chai, Bin Guo, Benjamin Yao, Chenlei Guo, Lili Mou. *VEG: Verbal ϵ -greedy for Semantic Exploration in Multi-Turn RL Agents*, Association for Computational Linguistics (ACL), 2026.
3. Kailash Gogineni, **Yongsheng Mei**, Karthikeya Gogineni, Peng Wei, Tian Lan, Guru Venkataramani. *RACER: A Caching Layer based Optimization for Accelerating Reinforcement Learning Workloads*, IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS), 2026.
4. Egemen Erbayat, **Yongsheng Mei**, Gina Adam, Suresh Subramaniam, Sean Coffey, Nathaniel D Bastian, Tian Lan. *LAMPS: A Learning-based Mobility Planning via Posterior State Inference using Gaussian Cox Process Models*, IEEE International Conference on Computer Communications (INFOCOM) Deep Learning for Wireless Communications, Sensing, and Security workshop, 2025.
5. **Yongsheng Mei**. *Decision-Making via Optimization in Challenging Environment: Bayesian and Multi-Agent Reinforcement Learning Perspectives*, The George Washington University Dissertations, 2025.
6. **Yongsheng Mei**, Mahdi Imani, Tian Lan. *Bayesian Optimization through Gaussian Cox Process Models for Spatio-temporal Data*, International Conference on Learning Representations (ICLR), 2024.
7. **Yongsheng Mei**, Hanhan Zhou, Tian Lan. *Projection-Optimal Monotonic Value Function Factorization in Multi-Agent Reinforcement Learning*, International Conference on Autonomous Agents and Multiagent Systems (AAMAS), 2024.
8. Yurong Chen, **Yongsheng Mei**, Tian Lan, Guru Venkataramani. *Exploring Effective Fuzzing Strategies to Analyze Communication Protocols*, ACM Digital Threats: Research and Practice, 2024.
9. Jingdi Chen, Hanhan Zhou, **Yongsheng Mei**, Gina Adam, Nathaniel Bastian, Tian Lan. *Real-time Network Intrusion Detection via Decision Transformers*, Association for the Advancement of Artificial Intelligence (AAAI) Artificial Intelligence for Cyber Security workshop, 2024.
10. Jiayu Chen, Bhargav Ganguly, Yang Xu, **Yongsheng Mei**, Tian Lan, Vaneet Aggarwal. *Deep Generative Models for Offline Policy Learning: Tutorial, Survey, and Perspectives on Future Directions*, Transactions on Machine Learning Research (TMLR), 2024
11. **Yongsheng Mei**, Tian Lan, Mahdi Imani, Suresh Subramaniam. *A Bayesian Optimization Framework for Finding Local Optima in Expensive Multi-Modal Functions*, European Conference on Artificial Intelligence (ECAI), 2023.
12. **Yongsheng Mei**, Tian Lan, Guru Venkataramani. *Exploiting Partial Common Information Microstructure for Multi-Modal Brain Tumor Segmentation*, International Conference on Machine Learning (ICML) Machine Learning for Multimodal Healthcare Data workshop, 2023.
13. Kailash Gogineni, **Yongsheng Mei**, Peng Wei, Tian Lan, Guru Venkataramani. *AccMER: Accelerating Multi-agent Experience Replay with Cache Locality-aware Prioritization*, IEEE International Conference on Application-specific Systems, Architectures and Processors (ASAP), 2023.
14. **Yongsheng Mei**, Hanhan Zhou, Tian Lan, Guru Venkataramani, Peng Wei. *MAC-PO: Multi-Agent Experience Replay via Collective Priority Optimization*, International Conference on Autonomous Agents and Multiagent Systems (AAMAS), 2023.
15. Kailash Gogineni, **Yongsheng Mei**, Guru Venkataramani, Tian Lan. *Verify-Pro: A Framework for Server Authentication using Communication Protocol Dialects*, IEEE Military Communications Conference (MILCOM), 2022.
16. **Yongsheng Mei**, Kailash Gogineni, Tian Lan, Guru Venkataramani. *MPD: Moving Target Defense through Communication Protocol Dialects*, International Conference on Security and Privacy in Communication Networks (SecureComm), 2021.
17. Hongfa Xue, **Yongsheng Mei**, Kailash Gogineni, Guru Venkataramani, Tian Lan. *Twin-Finder: Integrated Reasoning Engine for Pointer-related Code Clone Detection*, International Workshop on Software Clones (IWSC), 2020.

AWARD

2025 GW ECE Best Dissertation Honorable Mention Award

2023 European Conference on Artificial Intelligence *Call to Arms* Award

2020 – 2023 GW University Fellowship

2019 HUST Outstanding Graduates